GITAM Deemed to be University

Electronics and Communication Engineering | Course Code: EC305

Electromagnetic Waves & Transmission Lines - Comprehensive Syllabus Lect

Syllabus Focus: EC305 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Electromagnetic Waves & Transmission Lines. I

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Maxwell Equations & Electromagnetic Statics

" Syllabus Scope: Gauss, Faraday, and Ampere laws in differential and integral forms, Displacement curren

Unit 2: Uniform Plane Wave Propagation

" Syllabus Scope: Helmholtz wave equation, Propagation constant, Attenuation and phase constants, Intrins

EC305: Electromagnetic Waves & Transmission Lines - Advanced Topics & Assessment

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Reflection & Refraction at Media Boundaries

" Normal and oblique incidence, Snell law, Fresnel reflection and transmission coefficients, Brewster angle, T

Unit 4: Transmission Line Equations

" Transmission line differential equations, Characteristic impedance Z0, Propagation constant, Reflection coef

Unit 5: Smith Chart & Impedance Matching

" Smith chart coordinate geometry, Input impedance computation, Quarter-wave transformers, Single and double s

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the wave equation in a conducting medium from Maxwell equations and obtain expressions for alph
2. Using a Smith chart, design a single-stub parallel tuner to match a load of Z_L = 100 + j80 ohms to a
3. State and prove Poynting Theorem for electromagnetic energy conservation in a finite volume.

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Matthew N.O. Sadiku - Elements of Electromagnetics Oxford University Press 7th Ed
* E.C. Jordan, K.G. Balmain - Electromagnetic Waves and Radiating Systems Prentice Hall

Authorized NoteVault Educational Document " Verified for Registered University Students